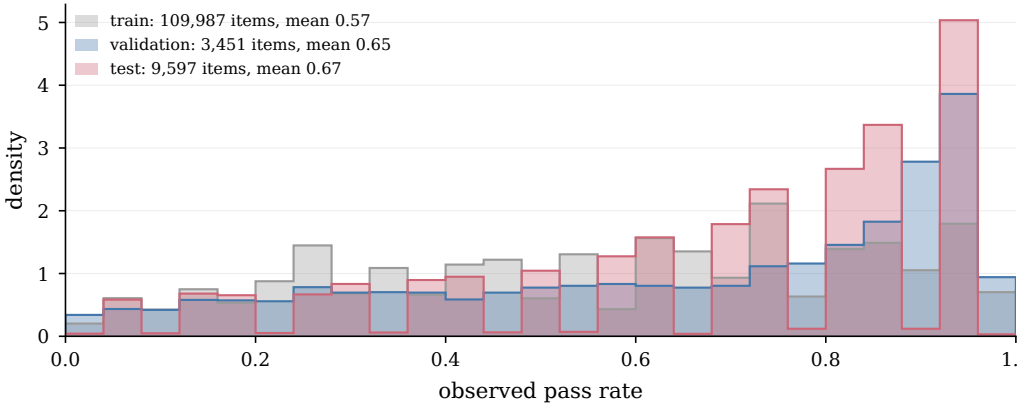
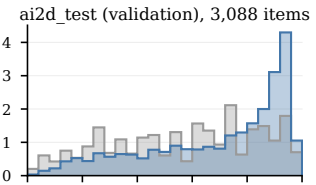
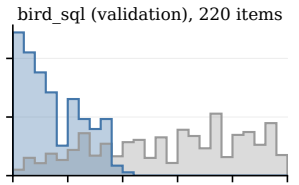
## Observed item difficulty by split

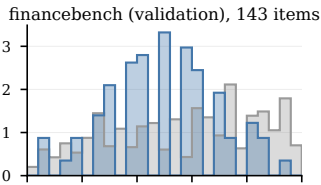
Pass rate = share of subjects answering the item correctly; items with ≥5 responses. Lower means harder.

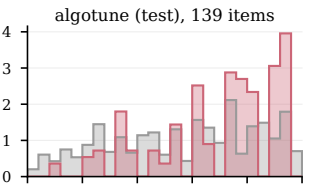


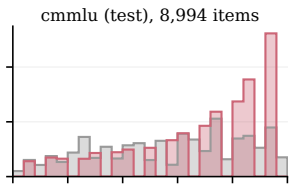

Each held-out benchmark against the pooled training distribution (grey).

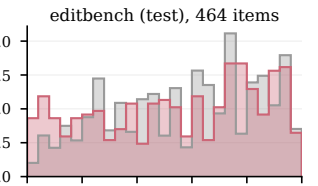


# Does the model's item belief track observed difficulty?

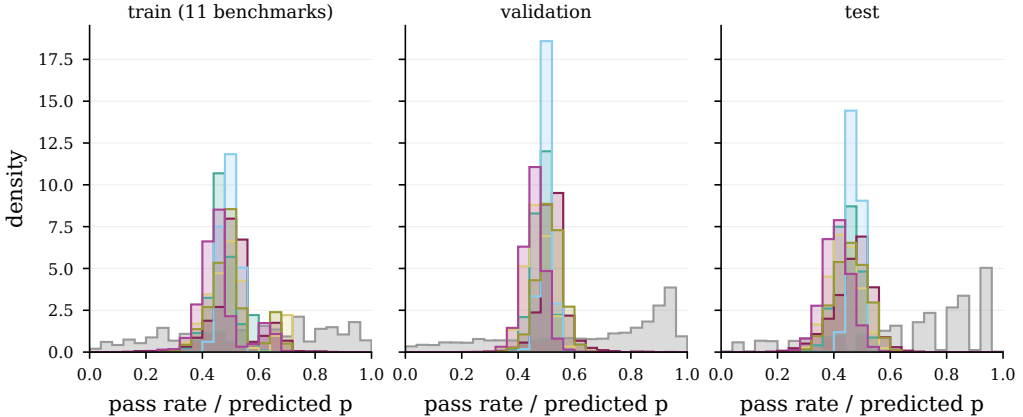
Budget-0 prediction (empty context) vs observed pass rate, per item. r Pearson, ρ Spearman, sd = spread.

| model | split | items | r | ρ | mean pred | mean obs | sd pred | sd obs |
|---|---|---|---|---|---|---|---|---|
| ANP | train | 109,987 | +0.186 | +0.172 | 0.475 | 0.570 | 0.050 | 0.267 |
| ANP | validation | 3,451 | +0.065 | +0.050 | 0.483 | 0.651 | 0.031 | 0.280 |
| ANP | test | 9,597 | -0.050 | -0.053 | 0.448 | 0.673 | 0.041 | 0.253 |
| ANP, budget-ladder training | train | 109,987 | +0.124 | +0.075 | 0.498 | 0.570 | 0.088 | 0.267 |
| ANP, budget-ladder training | validation | 3,451 | -0.060 | -0.070 | 0.466 | 0.651 | 0.044 | 0.280 |
| ANP, budget-ladder training | test | 9,597 | -0.061 | -0.060 | 0.436 | 0.673 | 0.053 | 0.253 |
| ANP + SAM | train | 109,987 | +0.178 | +0.160 | 0.516 | 0.570 | 0.081 | 0.267 |
| ANP + SAM | validation | 3,451 | +0.027 | +0.038 | 0.522 | 0.651 | 0.050 | 0.280 |
| ANP + SAM | test | 9,597 | -0.079 | -0.083 | 0.467 | 0.673 | 0.069 | 0.253 |
| ANP + prior anchor | train | 109,987 | +0.084 | +0.050 | 0.497 | 0.570 | 0.032 | 0.267 |
| ANP + prior anchor | validation | 3,451 | +0.140 | +0.069 | 0.499 | 0.651 | 0.018 | 0.280 |
| ANP + prior anchor | test | 9,597 | -0.051 | -0.059 | 0.474 | 0.673 | 0.021 | 0.253 |
| ANP + SAM, budget-ladder training | train | 109,987 | +0.185 | +0.163 | 0.501 | 0.570 | 0.079 | 0.267 |
| ANP + SAM, budget-ladder training | validation | 3,451 | +0.032 | +0.035 | 0.511 | 0.651 | 0.044 | 0.280 |
| ANP + SAM, budget-ladder training | test | 9,597 | -0.083 | -0.082 | 0.460 | 0.673 | 0.058 | 0.253 |
| ANP, prefix training | train | 109,987 | +0.191 | +0.162 | 0.461 | 0.570 | 0.078 | 0.267 |
| ANP, prefix training | validation | 3,451 | +0.071 | +0.058 | 0.455 | 0.651 | 0.038 | 0.280 |
| ANP, prefix training | test | 9,597 | -0.053 | -0.055 | 0.410 | 0.673 | 0.050 | 0.253 |
| *ANP, random-subject split* | *train* | *109,987* | *+0.310* | *+0.303* | *0.555* | *0.570* | *0.157* | *0.267* |
| *ANP, random-subject split* | *validation* | *3,451* | *+0.606* | *+0.640* | *0.632* | *0.651* | *0.187* | *0.280* |
| *ANP, random-subject split* | *test* | *9,597* | *+0.039* | *+0.023* | *0.546* | *0.673* | *0.164* | *0.253* |
| *ANP, random-subject split (width 12* | *train* | *109,987* | *+0.271* | *+0.284* | *0.467* | *0.570* | *0.114* | *0.267* |
| *ANP, random-subject split (width 12* | *validation* | *3,451* | *+0.604* | *+0.610* | *0.628* | *0.651* | *0.118* | *0.280* |
| *ANP, random-subject split (width 12* | *test* | *9,597* | *-0.009* | *-0.010* | *0.487* | *0.673* | *0.108* | *0.253* |

A model that learned transferable difficulty would keep r and its spread on validation and test; one that memorised its training benchmarks keeps them only on train. Italic rows: random-subject split, which also trained on ai2d_test and bird_sql (validation) and editbench (test); see the seen-vs-unseen page.

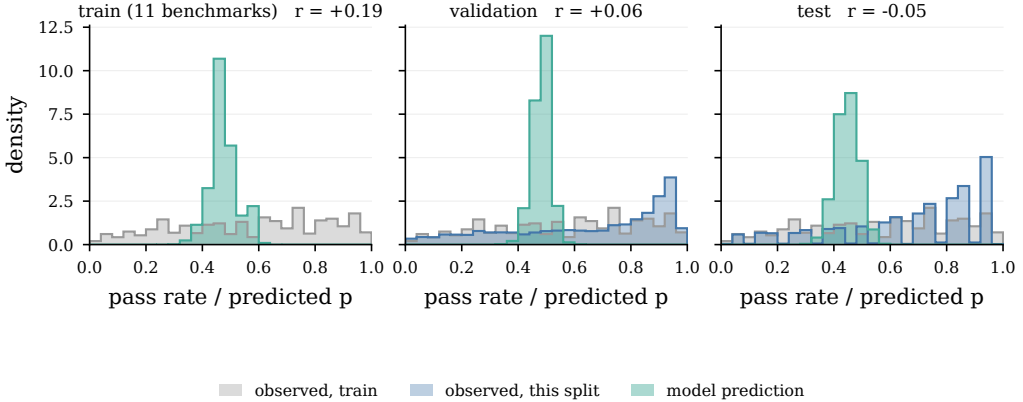
## All holdout models' item beliefs, by split

Grey: observed pass rate on that split. Colours: each model's budget-0 prediction, filled at low opacity.



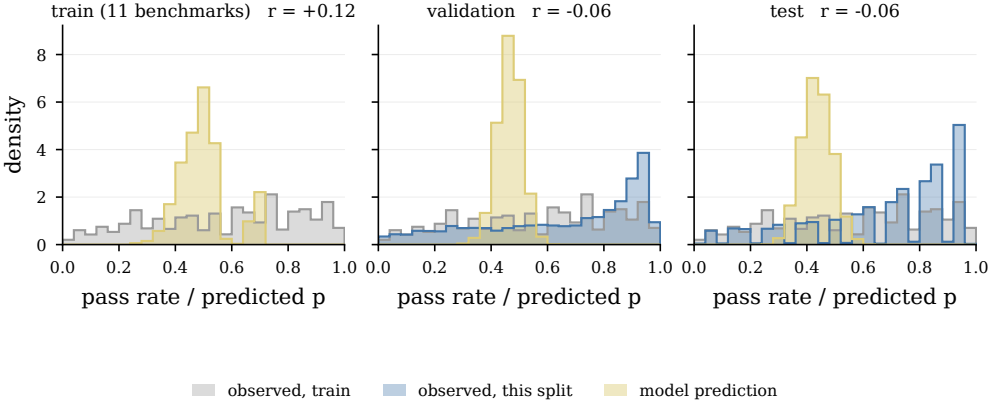
**ANP: predicted vs observed item difficulty**

Grey: train obs. Blue: this split obs. Colour: budget-0 prediction.



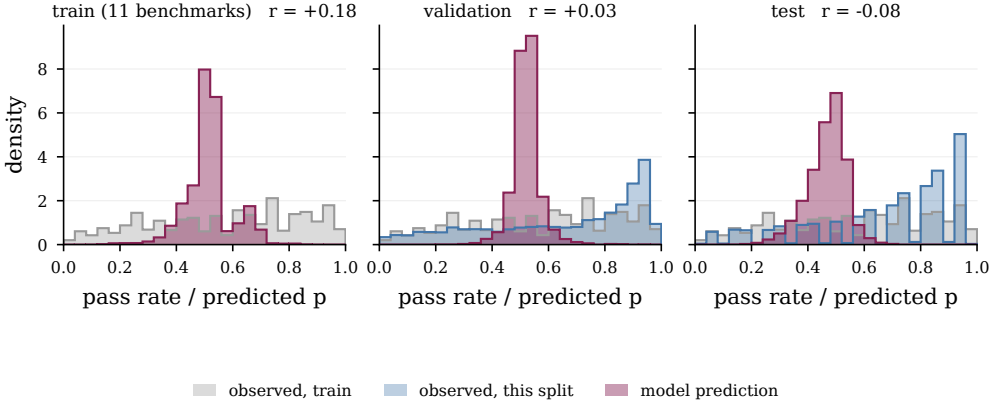
**ANP, budget-ladder training: predicted vs observed item difficulty**

Grey: train obs. Blue: this split obs. Colour: budget-0 prediction.



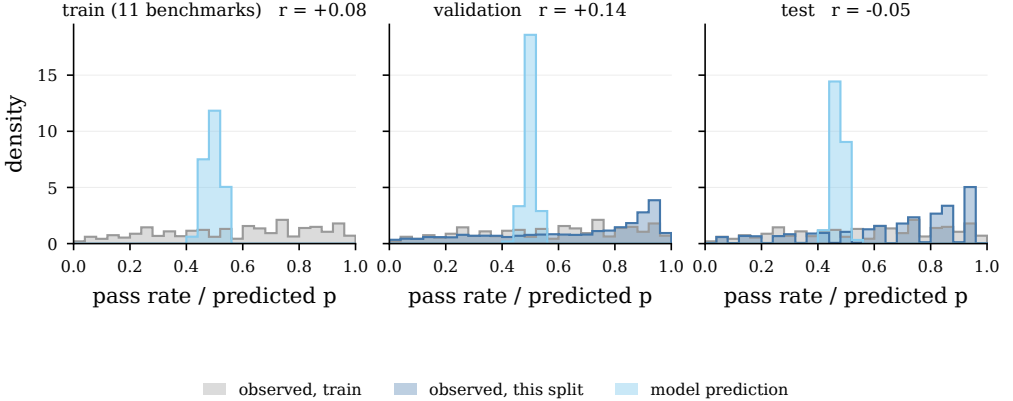
**ANP + SAM: predicted vs observed item difficulty**

Grey: train obs. Blue: this split obs. Colour: budget-0 prediction.



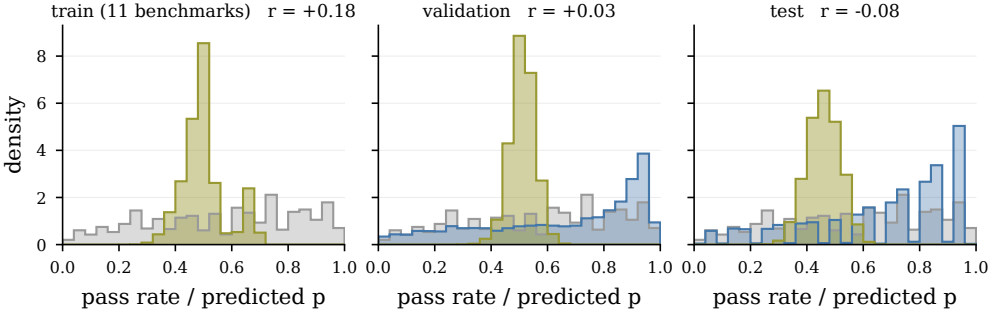
**ANP + prior anchor: predicted vs observed item difficulty**

Grey: train obs. Blue: this split obs. Colour: budget-0 prediction.



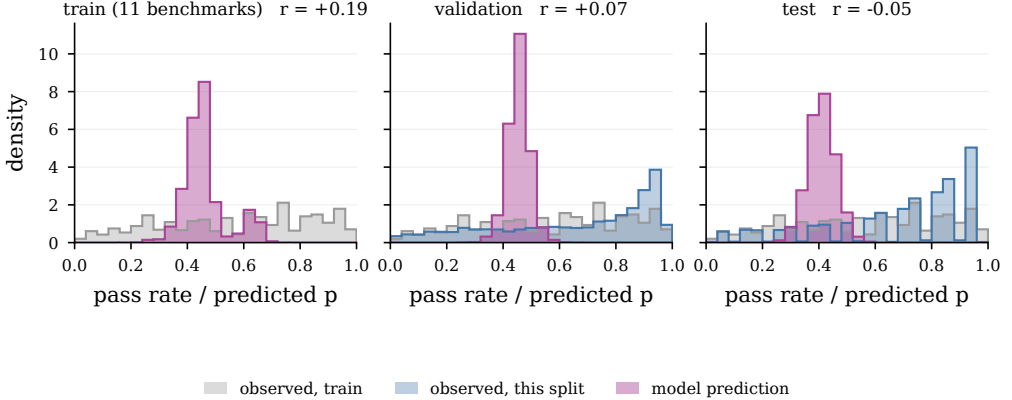
**ANP + SAM, budget-ladder training: predicted vs observed item difficulty**

Grey: train obs. Blue: this split obs. Colour: budget-0 prediction.



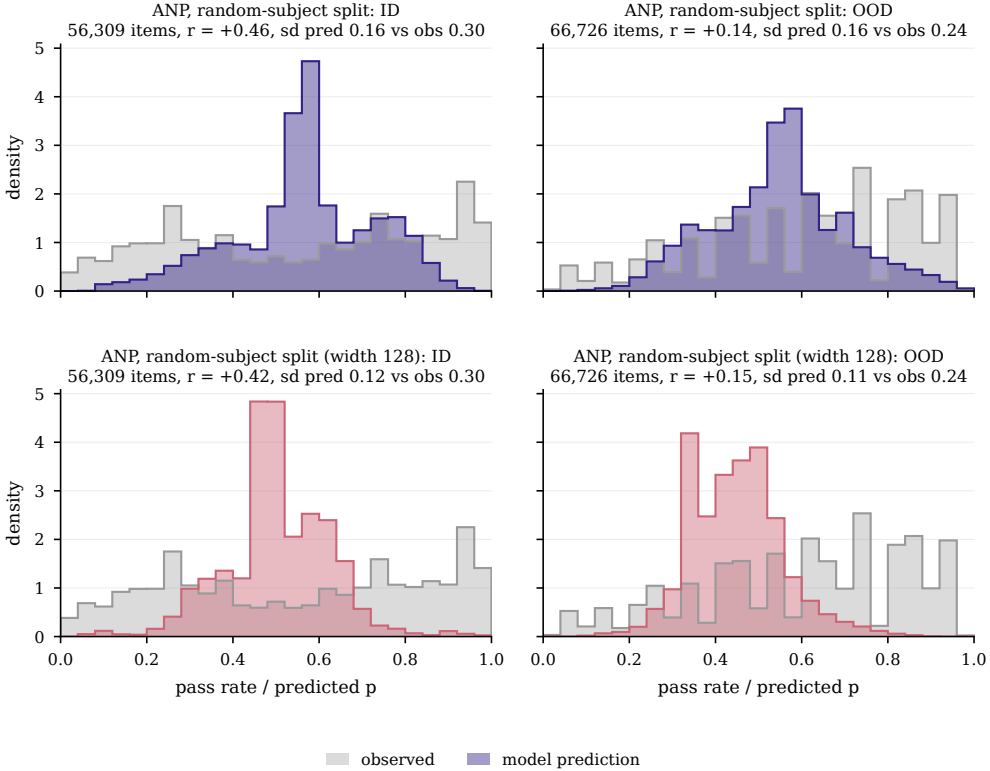
**ANP, prefix training: predicted vs observed item difficulty**

Grey: train obs. Blue: this split obs. Colour: budget-0 prediction.



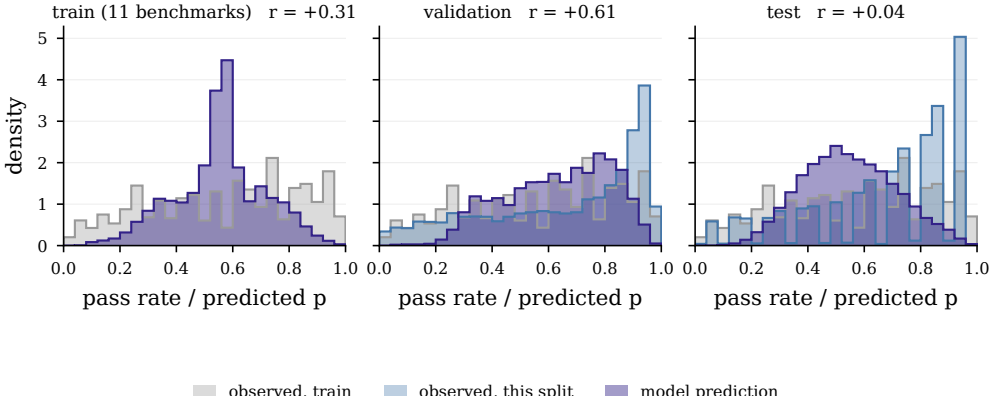
## Random-subject-split models: ID vs OOD benchmarks

Grey: observed. Colour: budget-0 prediction. ID = benchmark among the 7 trained on; OOD = any other.



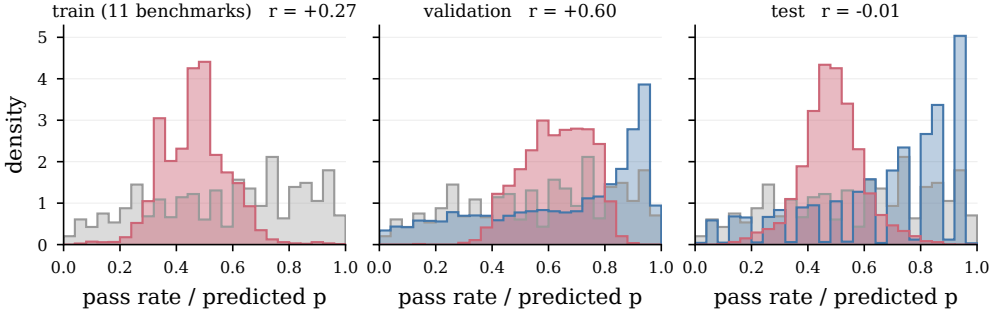
# ANP, random-subject split: predicted vs observed item difficulty

Random-subject split: trained on 7 benchmarks incl. ai2d_test, bird_sql, editbench. Grey: train obs. Blue: this split obs. Co



**ANP, random-subject split (width 128): predicted vs observed item difficulty**

Random-subject split: trained on 7 benchmarks incl. ai2d_test, bird_sql, editbench. Grey: train obs. Blue: this split obs. Co



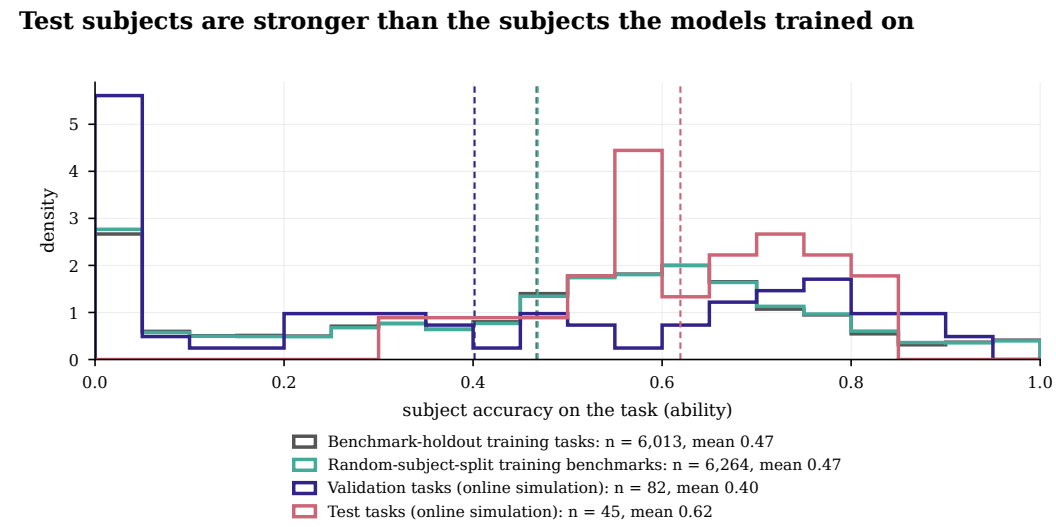



Figure. Distribution of subject ability, measured as each subject's accuracy on one benchmark (a task); dashed lines mark the means. The benchmark-holdout models trained on the grey tasks; the random-subject-split models trained on the seven benchmarks in teal, with subjects split at random. Validation and test tasks are those of the online simulation, scored on their evaluation pools. Test subjects are on average about 0.15 more accurate than the training subjects (0.62 against 0.47), and validation subjects slightly less (0.40), so a prior centred on the training mean starts too low on test and too high on validation. None of the validation or test subjects appear in training.

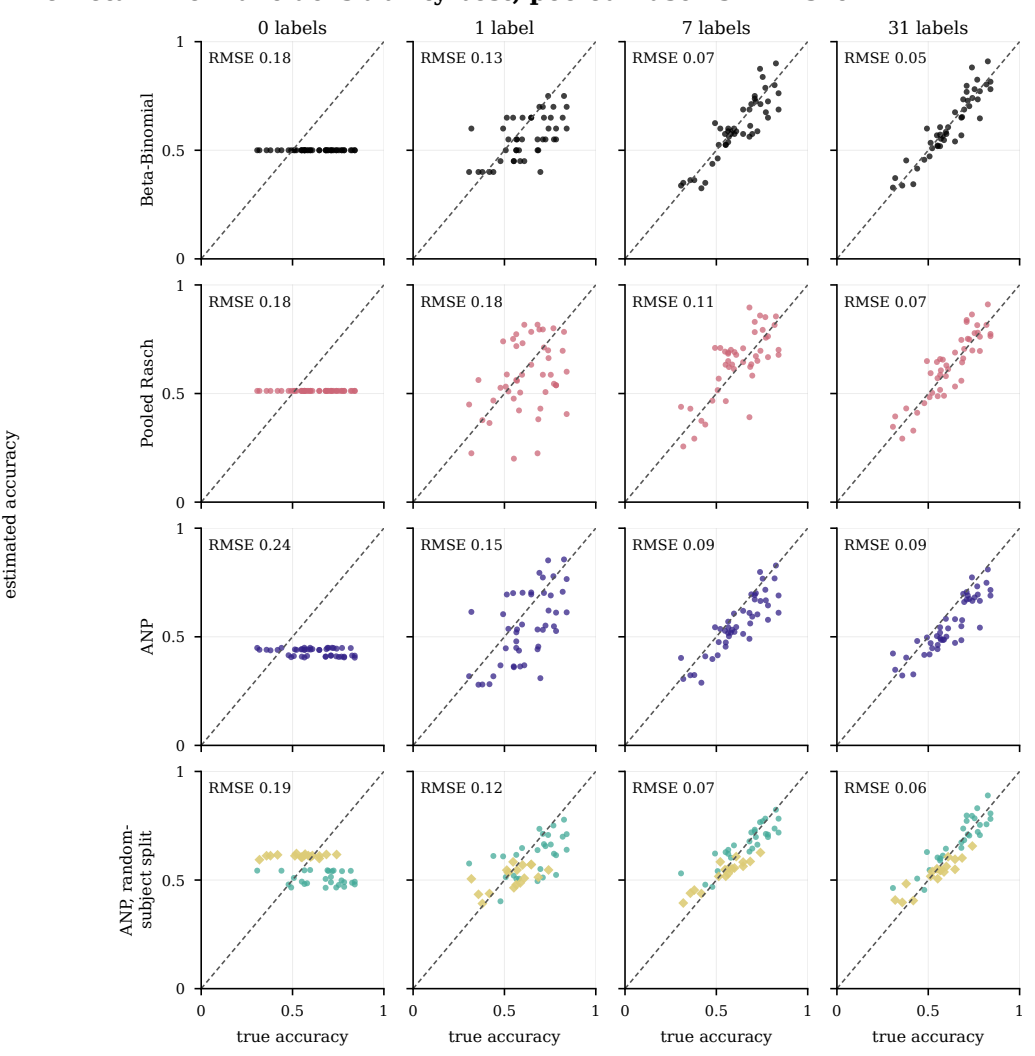



Figure (test). Each point is one test task: its true accuracy on the scored items (x) against the model's estimate of it (y), the mean probability the model assigns over those items, after 0, 1, 7 and 31 labels (random acquisition, averaged over candidate orders; pooled Rasch over 40 multi-pair draws). The dashed diagonal is a perfect estimate and each panel gives the RMSE. With no labels every model predicts nearly one level for everyone. The Beta-Binomial's estimate tracks the truth most closely from one label on; the random-subject-split ANP matches it, and the benchmark-holdout ANP stalls at an error of about 0.09 from seven labels on. Pooled Rasch shrinks ability harder (RMSE 0.18 after one label, 0.07 after 31): its lower Brier comes from pooling the benchmark rate and item difficulty, not from a sharper ability estimate. In the random-subject-split row, diamonds are editbench tasks, a benchmark that model trained on; circles are algotune and cmmlu, which it never saw.

## On validation the ANP tracks ability about as well as the Beta-Binomial

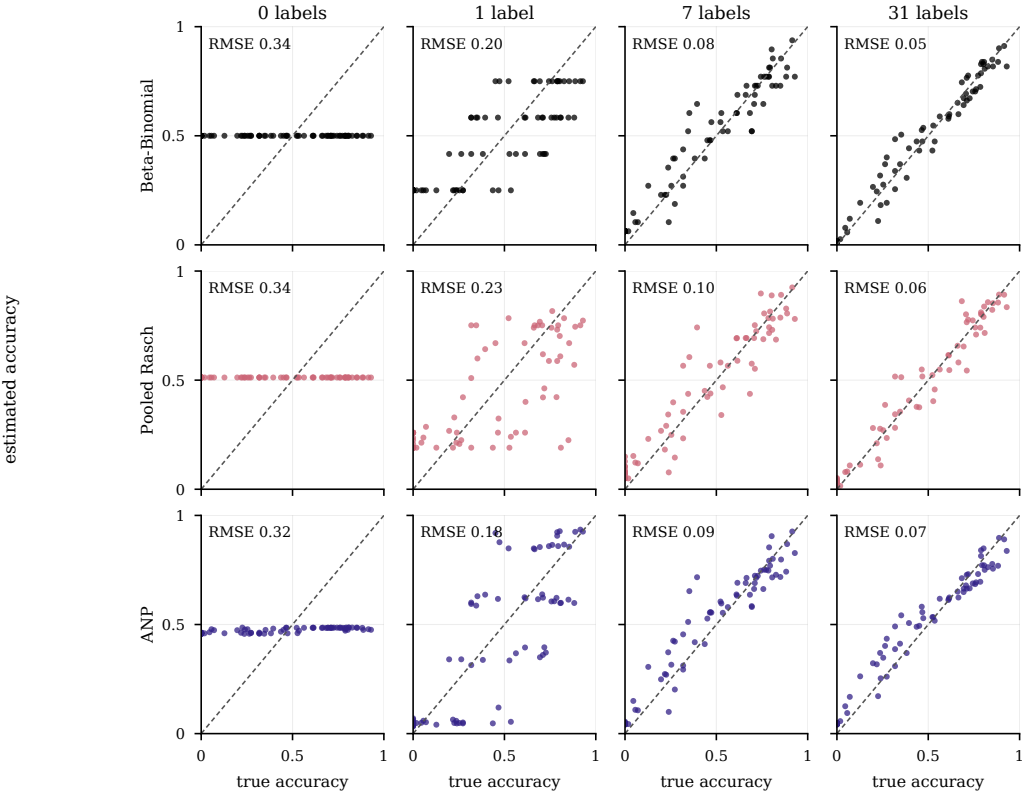



Figure (validation). Each point is one validation task: its true accuracy on the scored items (x) against the model's estimate of it (y), the mean probability the model assigns over those items, after 0, 1, 7 and 31 labels (random acquisition, averaged over candidate orders; pooled Rasch over 40 multi-pair draws). The dashed diagonal is a perfect estimate and each panel gives the RMSE. With no labels every model predicts nearly one level for everyone. The ANP's estimate is slightly closer than the Beta-Binomial's at one and three labels and slightly further from seven on; pooled Rasch again shrinks ability the most. Pooled Rasch is shown for the 72 validation tasks its 40 draws covered.